A SYNOPSIS

ON

"REAL-TIME UPI FRAUD DETECTION USING MACHINE LEARNING"

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ABSTRACT:

The increasing adoption of Unified Payments Interface (UPI) has transformed India into one of the world's fastest-growing digital economies. However, this growth is coupled with a surge in fraudulent activities, such as phishing, unauthorized access, and fake payment requests. Traditional rule-based fraud detection mechanisms are unable to keep pace with evolving fraud strategies, leaving users vulnerable.

The proposed mini-project, "Real-Time UPI Fraud Detection using Machine Learning", aims to create a fraud detection system capable of monitoring transactions in real time. The system will use machine learning models to analyse patterns of normal transactions versus suspicious ones. A new feature is the integration of behavioural biometrics (typing speed, device usage, swipe patterns) to differentiate between genuine and fraudulent users. This multi-layered approach enhances reliability and minimizes false positives.

In the current scenario, where India records over 14 billion monthly UPI transactions (2025), the system is highly useful to protect users, build trust in digital platforms, and support safe financial inclusion.

INTRODUCTION:

UPI has become the backbone of India's cashless economy, processing billions of transactions every month. Its ease of use has also attracted fraudsters who exploit user unawareness and weak monitoring systems. Cases of fraud such as fake UPI payment confirmations, phishing links, SIM swapping, and social engineering scams have become common.

Conventional fraud detection systems depend on fixed rules, which are unable to adapt to new attack methods. This results in either missed fraud cases or excessive false alarms. To address this, machine learning-based systems provide a dynamic, self-learning solution capable of identifying unknown fraud patterns in real time [Sharma et al., 2022].

The proposed project develops such a system, combining transaction analysis with user behaviour monitoring, making fraud detection more robust and scalable in today's fast-changing digital payment ecosystem.

PROBLEM STATEMENT:

Despite advanced payment systems, UPI still faces critical challenges in fraud detection:

- Delayed identification → money is lost before fraud is flagged.
- Inability to detect new fraud strategies, as rule-based systems cannot evolve [Gupta & Singh, 2021].

- High false positives, leading to unnecessary transaction blocks.
- Lack of user behaviour consideration, making it easier for attackers to impersonate genuine users.

Thus, there is a need for a real-time, adaptive fraud detection system that leverages machine learning and user behaviour analysis to ensure secure UPI transactions**.**

LITERATURE SURVEY:

Studies have shown that machine learning significantly improves fraud detection accuracy compared to rule-based methods.

- Supervised learning methods (Random Forest, Logistic Regression, Neural Networks) have been successful in classifying fraudulent vs. legitimate transactions [Sharma et al., 2022].
- Unsupervised methods (Isolation Forest, Autoencoders) are useful in detecting unknown fraud types [Bhattacharya, 2020].
- Research highlights that adding contextual features (location, device ID, time of transaction) improves fraud detection [Gupta & Singh, 2021].

However, most existing approaches do not integrate behavioural biometrics, which is emerging as a powerful tool for user authentication. This project introduces that layer, making the system more resistant to impersonation and social engineering fraud.

OBJECTIVES OF THE PROPOSED RESEARCH:

1. To analyse UPI transaction datasets to identify fraudulent patterns.
2. To develop hybrid machine learning models combining supervised and unsupervised learning for real-time fraud detection.
3. To integrate behavioural biometrics (typing speed, device interaction, swipe gestures) as an additional fraud prevention layer.
4. To design a real-time fraud alert system via a dashboard or mobile app.
5. To ensure secure and trustworthy UPI transactions that adapt to new fraud patterns.

METHODOLOGY:

1. Data Collection – Gather anonymized UPI transaction data from NPCI and open financial datasets, including transaction history, device details, and location [NPCI, 2023].

2. Data Preprocessing – Handle missing values, normalize data, and engineer features like transaction frequency, velocity, and behaviour patterns.

3. Exploratory Data Analysis (EDA) – Visualize fraud cases, detect correlations, and study seasonal trends.

4. Model Development –

    - Supervised ML (Random Forest, XGBoost, Neural Networks).

    - Unsupervised ML (Isolation Forest, Autoencoders) for unseen fraud.

    - Behavioural biometric analysis for user-device patterns.

5. Real-Time System Design – Build a streaming fraud detection engine capable of sending instant alerts before transaction completion.

6. Testing & Validation – Evaluate models with precision, recall, F1-score, and false positive rate.

7. Deployment – Integrate with UPI platforms and design a farmer-friendly mobile interface for fraud alerts.

EXPECTED OUTCOME:

- A real-time UPI fraud detection system powered by machine learning.

- Behavioural biometric layer for enhanced fraud prevention (new contribution).

- Instant alerts to block suspicious transactions before completion.

- Reduction in financial losses and improved user confidence.

- Support for India's secure digital financial infrastructure.

SUMMARY:

The Real-Time UPI Fraud Detection using Machine Learning mini-project introduces a multi-layered fraud detection system that not only analyses transaction data but also incorporates behavioural biometrics for stronger security. This approach makes the system adaptable to emerging fraud strategies and highly useful in the current scenario of rising UPI fraud cases. By combining machine learning with real-time monitoring, the project contributes to a secure, reliable, and scalable digital payment ecosystem.

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